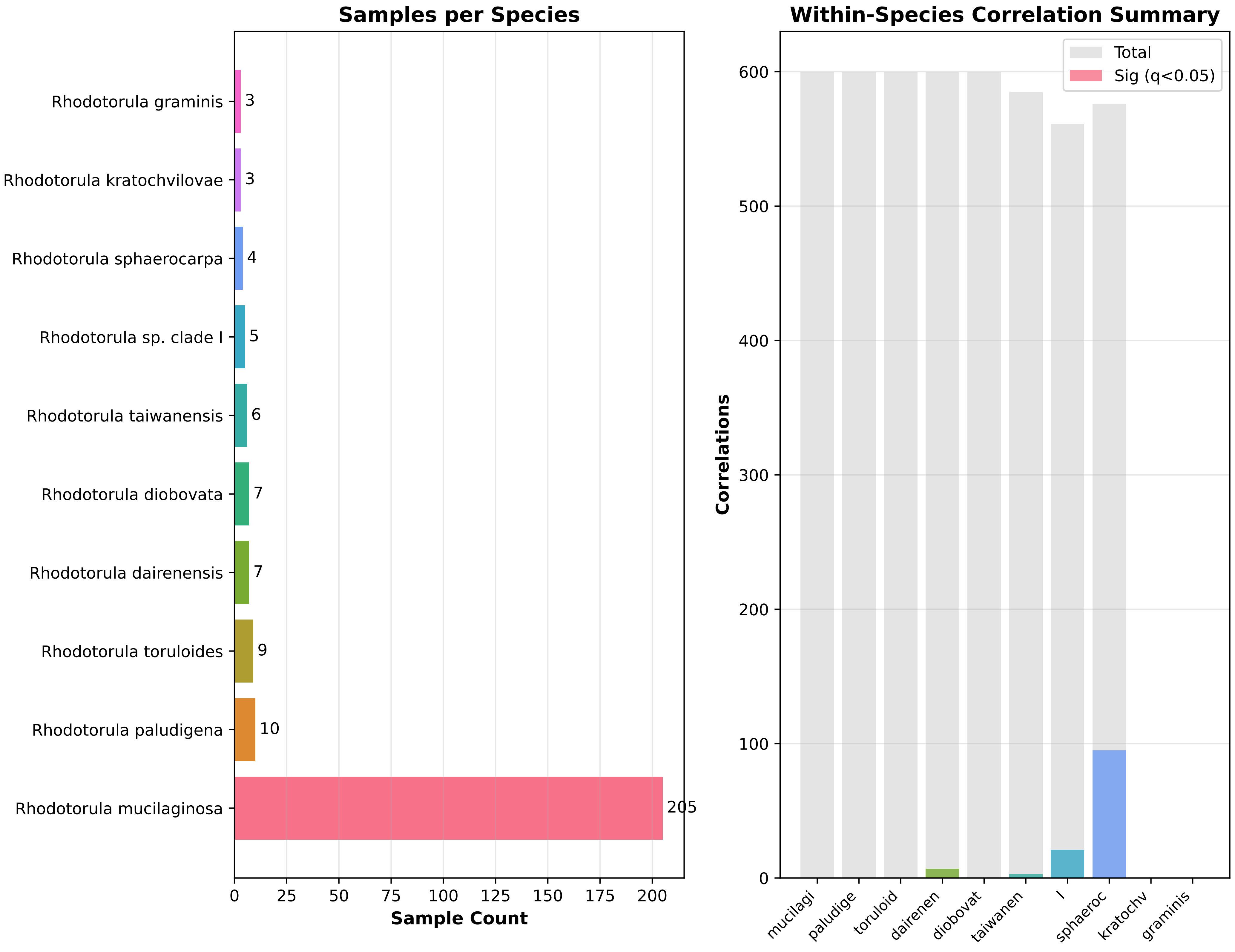
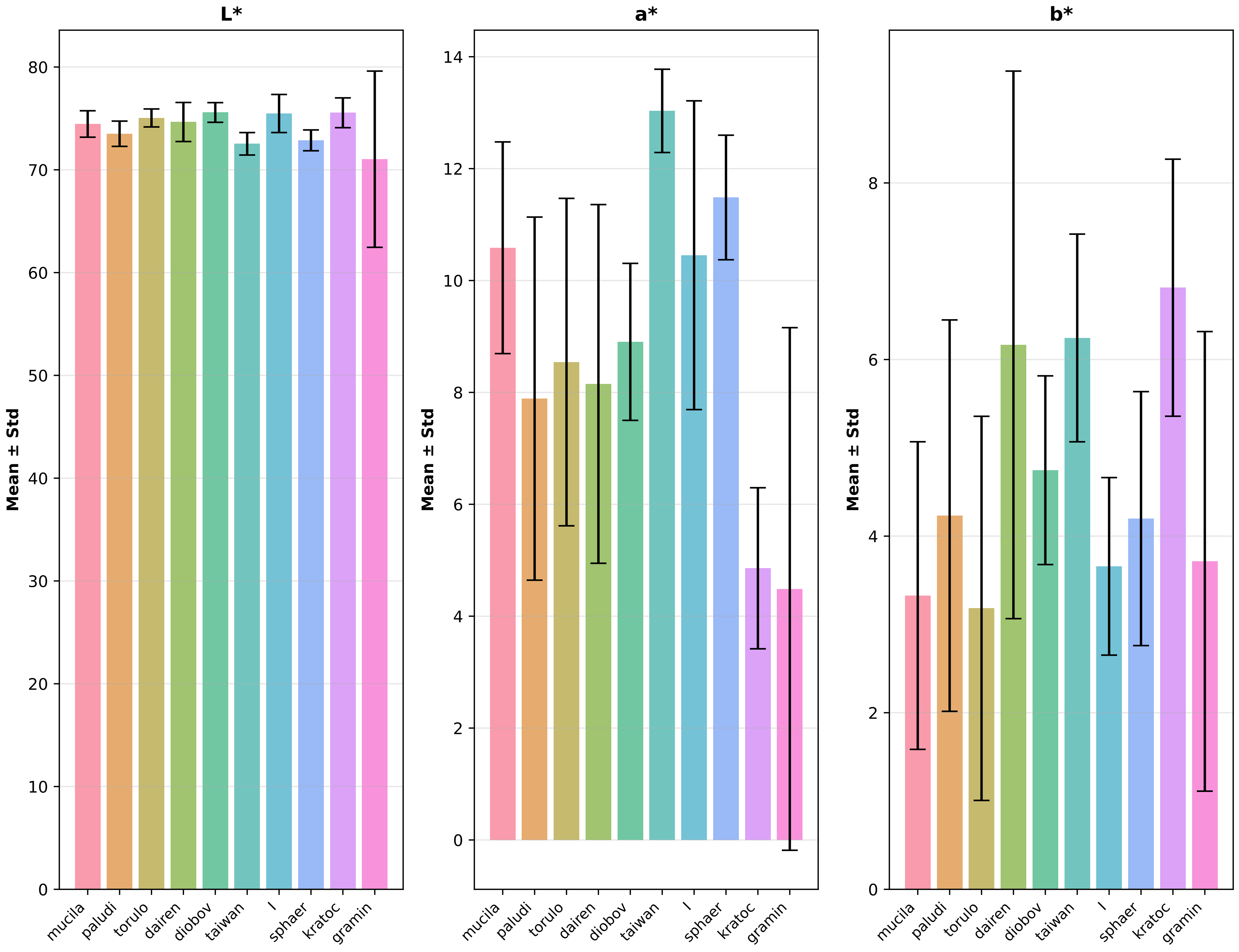
Stratified Species Analysis
Metabolite-Phenotype Correlations

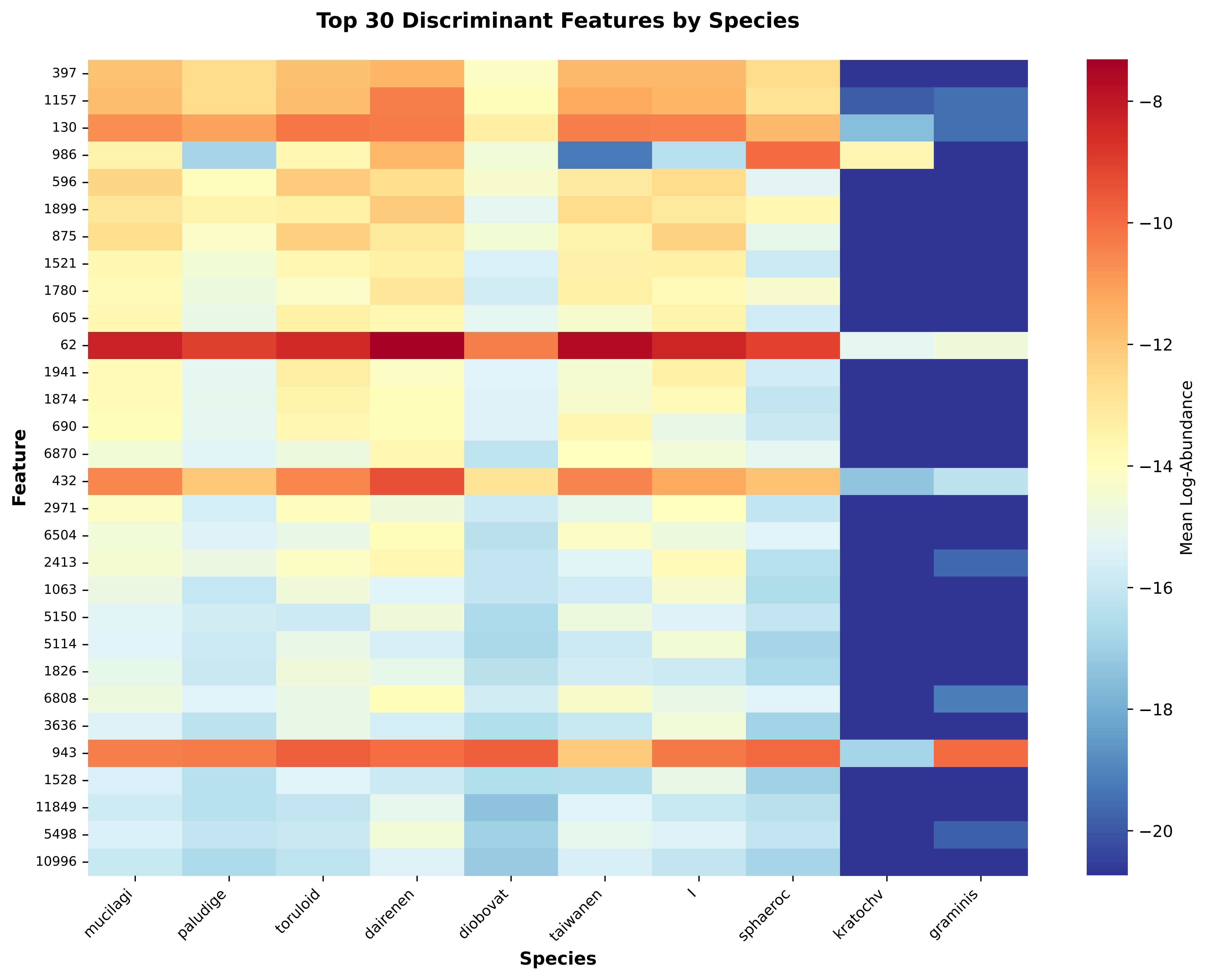
10 major species | 590 samples
7341 features | 3 phenotypes

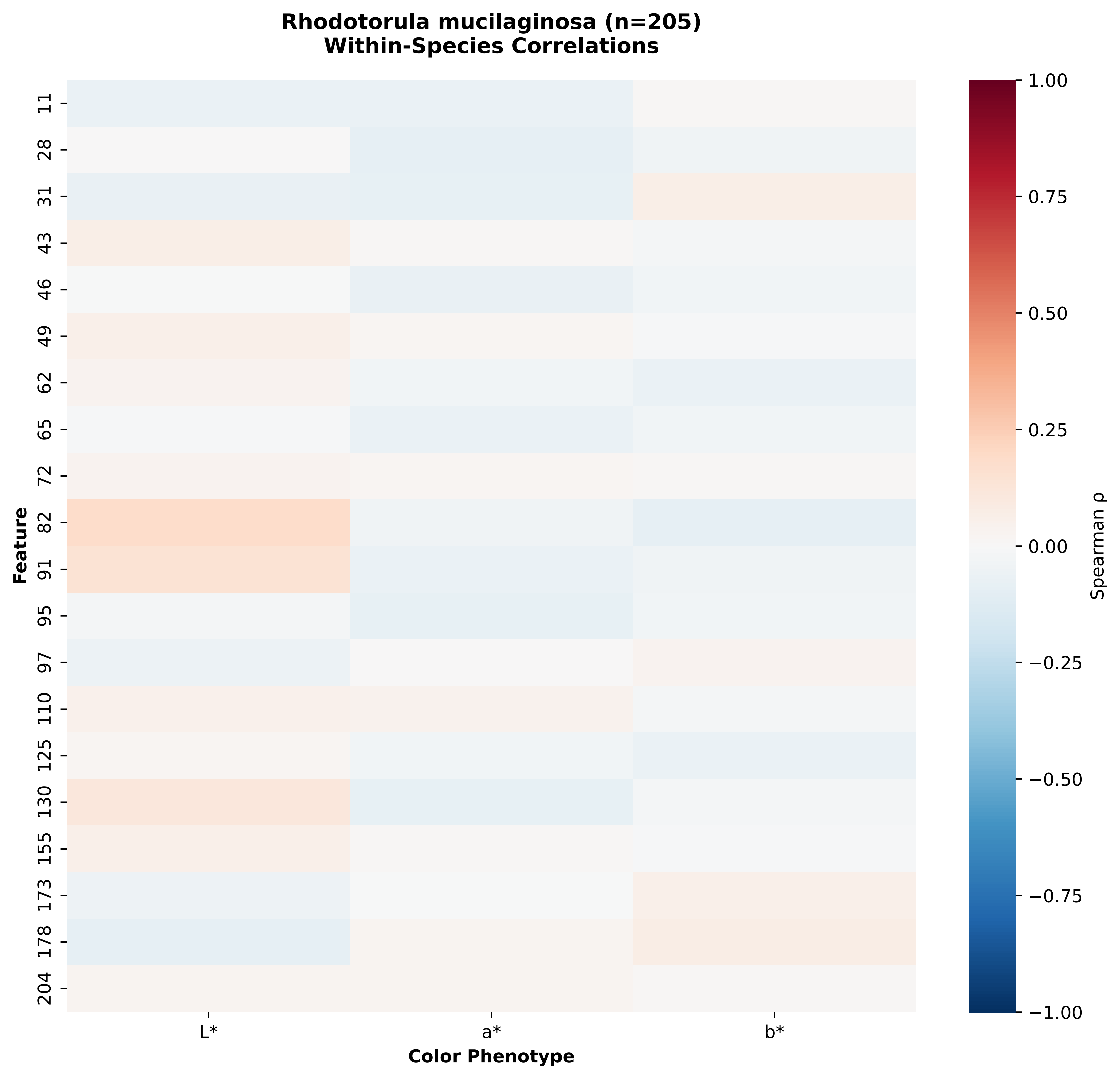
- Between-species discriminant features
- Phenotype variation across species
- Within-species correlations


Samples per Species
Rhodotorula graminis
Rhodotorula kratochvilovae
Rhodotorula sphaerocarpa
Rhodotorula sp. clade I
Rhodotorula taiwanensis
Rhodotorula diobovata
Rhodotorula dairenensis
Rhodotorula toruloides
Rhodotorula paludigena
Rhodotorula mucilaginosa
3
3
4
5
6
7
7
9
10
205
0 25 50 75 100 125 150 175 200
Sample Count
Within-Species Correlation Summary
Total
Sig (q<0.05)
Correlations
0 100 200 300 400 500 600
mucilagi
paludige
toruloid
dairenen
diobovat
taiwanen
I
sphaeroc
kratochv
graminis

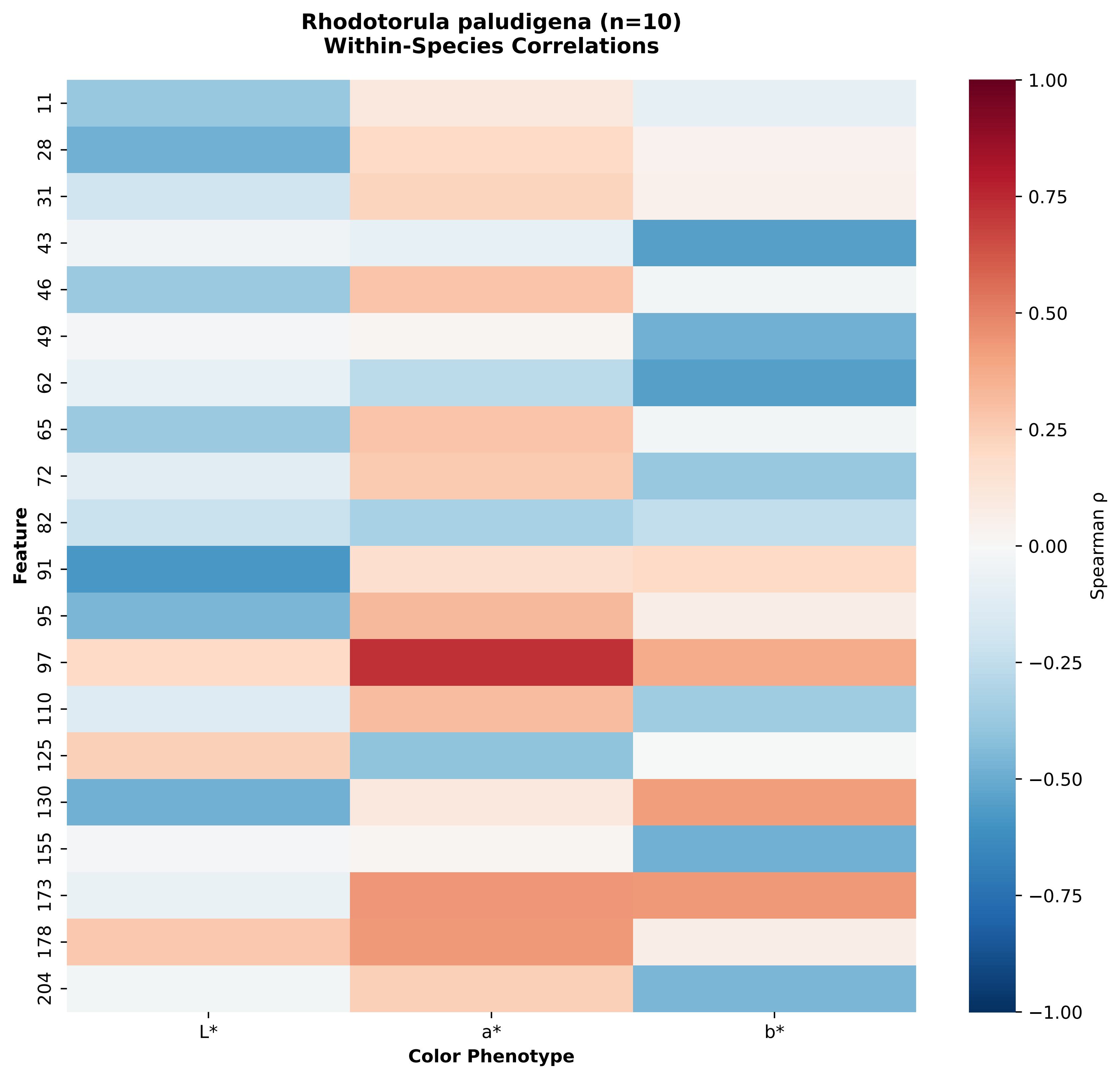


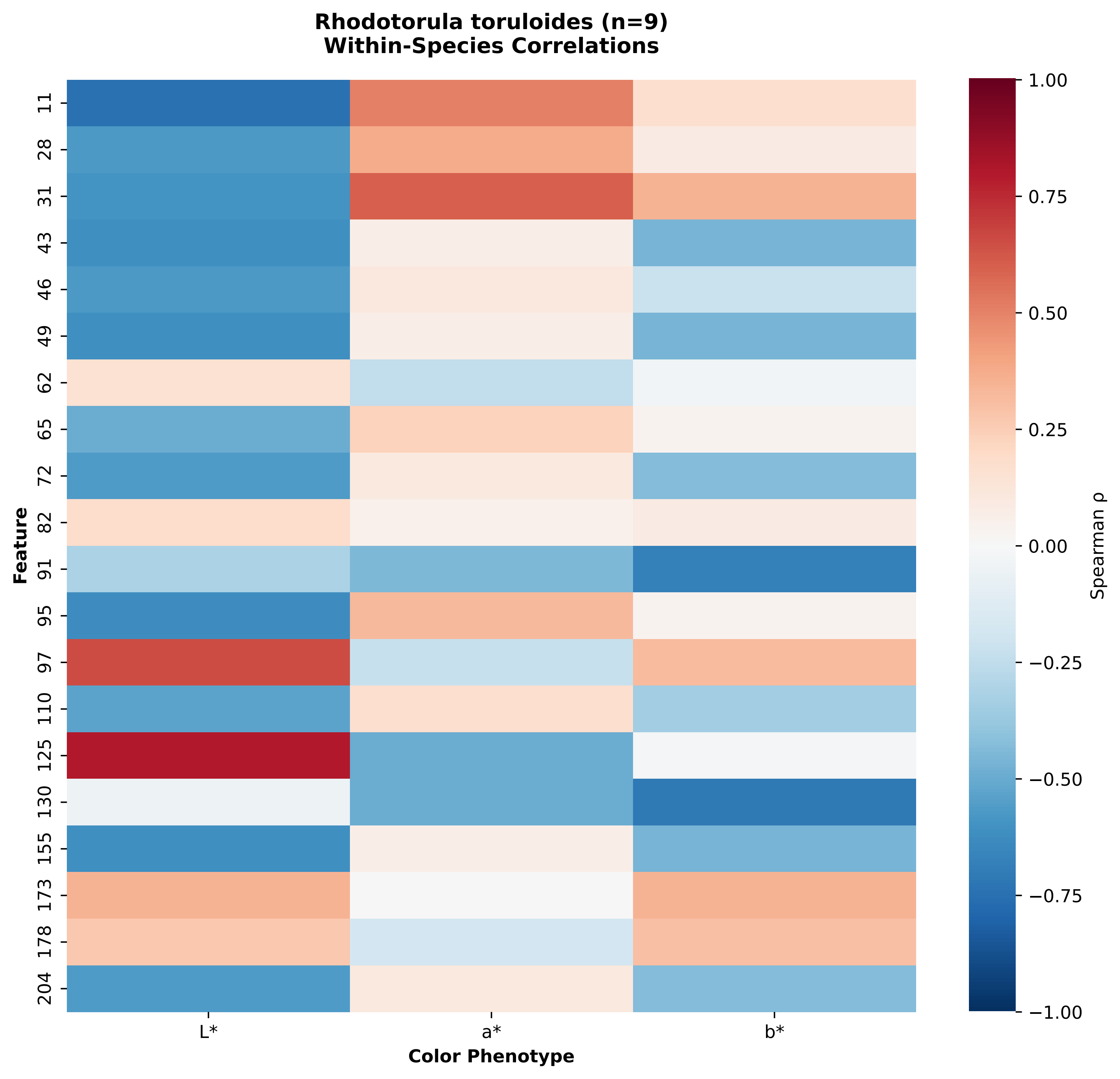
L*
a*
b*
Mean ± Std
mucila
paludi
torulo
dairen
diobov
taiwan
I
sphaer
kratoc
gramin

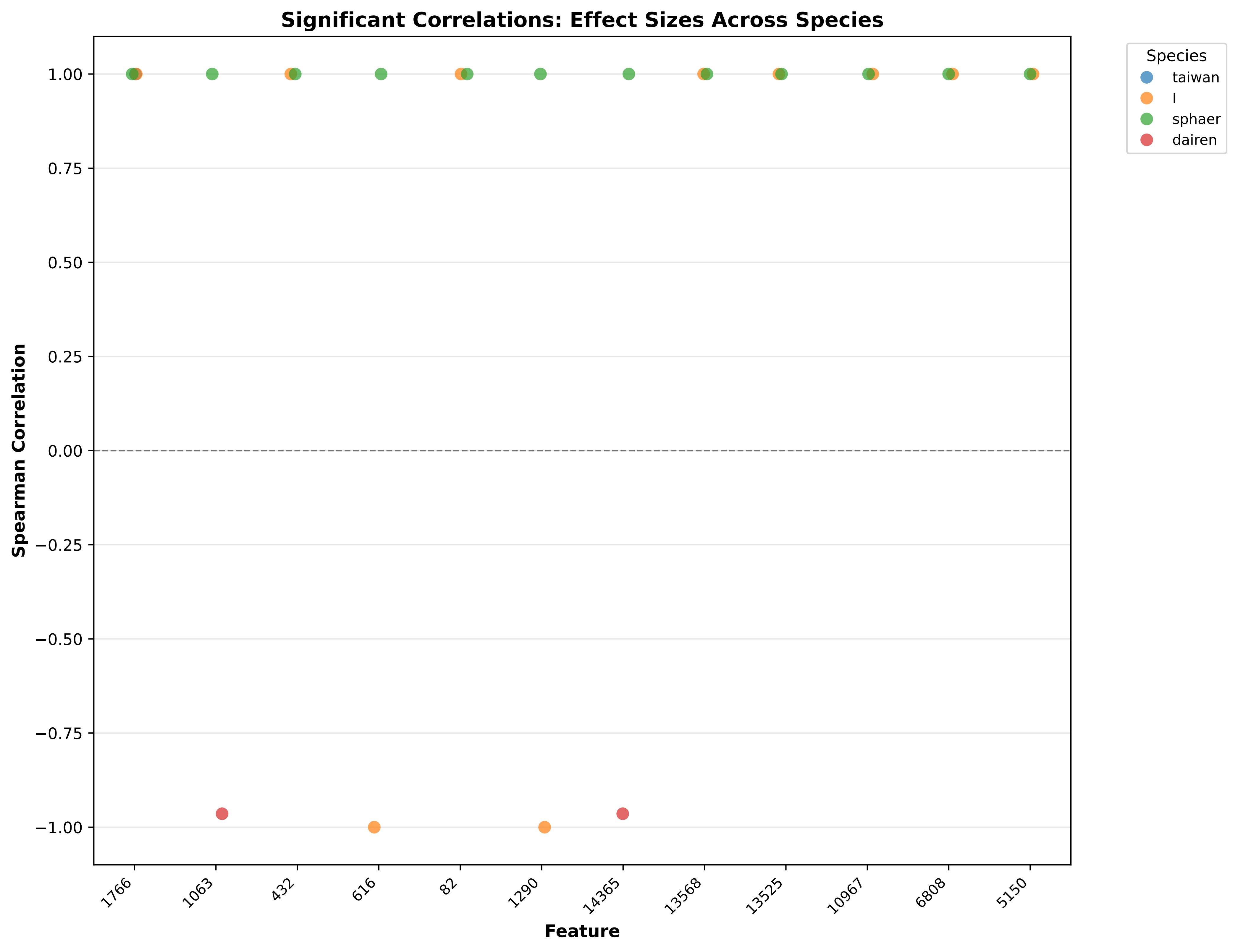

Top 30 Discriminant Features by Species
Feature
397
1157
130
986
596
1899
875
1521
1780
605
62
1941
1874
690
6870
432
2971
6504
2413
1063
5150
5114
1826
6808
3636
943
1528
11849
5498
10996
mucilagi
paludige
toruloid
dairenen
diobovat
taiwanen
l
sphaeroc
kratochv
graminis
Species
Mean Log-Abundance
−8
−10
−12
−14
−16
−18
−20



Rhodotorula mucilaginosa (n=205)
Within-Species Correlations
Feature
11
28
31
43
46
49
62
65
72
82
91
95
97
110
125
130
155
173
178
204
L*
a*
b*
Color Phenotype
Spearman ρ
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Rhodotorula paludigena (n=10)
Within-Species Correlations
Feature
11
28
31
43
46
49
62
65
72
82
91
95
97
110
125
130
155
173
178
204
L*
a*
b*
Color Phenotype
Spearman ρ
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Rhodotorula toruloides (n=9)
Within-Species Correlations
Feature
11
28
31
43
46
49
62
65
72
82
91
95
97
110
125
130
155
173
178
204
L*
a*
b*
Color Phenotype
Spearman ρ
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00



Significant Correlations: Effect Sizes Across Species
Spearman Correlation
Feature
Species
taiwan
I
sphaer
dairen
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
1766
1063
432
616
82
1290
14365
13568
13525
10967
6808
5150

Key Findings & Summary

BETWEEN-SPECIES SIGNALS (Discriminant Features):

- Identified 200 features with high variance across species
- Top feature: 397 (variance = 12.10)
- Species show distinct metabolite profiles and phenotypic variation

WITHIN-SPECIES CORRELATIONS:

- 4,722 feature-phenotype correlations tested across species
- 126 significant associations (FDR q<0.05)

Correlation strength by species:

- Rhodotorula mucilaginosa: 0/600 (0.0%)
- Rhodotorula paludigena: 0/600 (0.0%)
- Rhodotorula toruloides: 0/600 (0.0%)

BIOLOGICAL INTERPRETATION:

- Metabolite composition varies significantly between species (diversified metabolism)
- Within species, specific metabolites correlate with visible color phenotypes
- Correlation patterns vary by species, suggesting species-specific biosynthetic pathways
- Smaller species show stronger correlations (directional metabolite-phenotype associations)
- Largest species (R. mucilaginosa, n=205) shows weaker within-species signals